

Contents

PAN211x Register Reference	4
Address Space and Page Concept	4
I2C Access Byte Encoding	4
Page 0 Register Map	5
Page 0 Register Details	8
0x00 — PAGECFG — Page Bank Select	8
0x01 — TRXFIFO — TX/RX FIFO Access	8
0x02 — STATECFG — State Machine Control	8
0x03 — SYSCFG — System Control	9
0x04 — SPICFG — SPI / Bus Interface Config	10
0x05 — XTALCFG — Crystal Load-Capacitor Trim	10
0x06 — I2CCFG — I2C Interface Config	10
0x07 — WMODECFG0 — Work Mode Configuration 0	11
0x08 — WMODECFG1 — Work Mode Configuration 1	11
0x09 — RXPLENCFG — Receive Payload Length	12
0x0A — TXPLENCFG — Transmit Payload Length	12
0x0B — RFIRQCFG — Interrupt Mask	12
0x0C — PIDCFG — PID Identifier Configuration	13
0x0D — TRXTWTLCFG — TX/RX Switch Wait Time (Low)	13
0x0E — TRXTWTHCFG — TX/RX Switch Wait Time (High)	13
0x0F-0x13 — PIPE0RXADDR0-4 — Pipe 0 RX Address	13
0x14-0x18 — TXADDR0-4 — TX Destination Address	14
0x19 — PKTEXTCFG — Packet Extension Configuration	14
0x1A — WHITENCFG — Whitening Configuration	14
0x1B — TXHDR0CFG — TX Header Byte 0	15
0x1C — TXHDR1CFG — TX Header Byte 1	15
0x1D — TXRAMADDRCFG — TX FIFO RAM Start Address	15
0x1E — RXRAMADDRCFG — RX FIFO RAM Start Address	15
0x1F — RXPIPECFG — Multi-Pipe RX Enable	16
0x20-0x24 — PIPE1RXADDR0-4 — Pipe 1 RX Address	16
0x25-0x28 — PIPE2-5 RXADDR0 — Pipes 2-5 Address LSB	16
0x29 — TXAUTOCFG — Auto-Retransmit Configuration	16
0x2A — TRXMODECFG — TX/RX Mode Configuration	17
0x2B — RXTIMEOUTLCFG — RX Timeout (Low)	17
0x2C — RXTIMEOUTHCFG — RX Timeout (High)	17
0x2D — BLEMATCHCFG0 — BLE Filter Configuration	18
0x2E — BLEMATCHCFG1	18
0x2F-0x34 — WLIST0-5CFG — BLE Whitelist AdvA	18
0x35 — BLEMATCHSTARTCFG — BLE Filter Payload Start	18
0x36 — RFDATARATECFG — Air Data Rate	18
0x37-0x38 — Undocumented	19
0x39 — RFCHANNELCFG — RF Channel	19

0x3A-0x42 — RF Analog Tuning (Undocumented, Page 0)	19
0x43 — RF Driver Tuning (Undocumented, Page 0) ⚙	20
0x44 — RF Output Tuning (Undocumented, Page 0) ⚙	20
0x45 — IRQMUXCFG — IRQ Pin Multiplexing	20
0x46-0x54 — RF Analog Tuning (Undocumented, Page 0) ⚙	20
0x55-0x61 — RF Analog Tuning Bank (Undocumented, Page 0) ⚙	20
0x62-0x65 — RF Analog Tuning (Undocumented, Page 0) ⚙	21
0x66 — RF Analog Tuning (Undocumented, Page 0) ⚙	21
0x67 — RF Analog Tuning (Undocumented, Page 0) ⚙	21
0x68 — RF Analog Tuning (Undocumented, Page 0) ⚙	21
0x69-0x6D — RF Analog Tuning (Undocumented, Page 0) ⚙	21
0x6E — RF Analog Tuning (Undocumented, Page 0) ⚙	21
0x6F — MISCCFG — Miscellaneous Configuration	22
0x70-0x72 — Undocumented (Page 0)	22
0x73 — RFIRQFLG — RF Interrupt Status Flags	22
0x74 — STATUS0 — Status Register 0 (Read-Only)	23
0x75 — STATUS1 — Received Header Byte 0 (Read-Only)	23
0x76 — STATUS2 — Received Header Byte 1 (Read-Only)	23
0x77 — STATUS3 — Received Payload Length (Read-Only)	23
0x78-0x79 — Undocumented (Page 0)	24
0x7A — PKTRSSIL — Last Packet RSSI (Low)	24
0x7B — PKTRSSIH — Last Packet RSSI (High)	24
0x7C-0x7D — Undocumented (Page 0)	24
0x7E — RTRSSIL — Ambient Noise RSSI (Low)	24
0x7F — RTRSSIH — Ambient Noise RSSI (High)	24
Page 1 Register Details	24
0x04 (Page 1) — OTP Data Register ⚙	24
0x05 (Page 1) — OTP Mode Control ⚙	25
OTP Calibration Data Fields ⚙	25
0x1B (Page 1) — Calibration Control Register ⚙	25
0x27 (Page 1) — RF Analog Tuning ⚙	26
0x32-0x33 (Page 1) — RF Analog Tuning ⚙	26
0x37 (Page 1) — RF Analog Tuning ⚙	26
0x3A (Page 1) — RF Analog Tuning ⚙	26
0x3C (Page 1) — TX Power Amplitude Control ⚙	26
0x3E (Page 1) — RF Analog Tuning ⚙	26
0x3F (Page 1) — RF Analog Tuning ⚙ (Undocumented)	26
0x40 (Page 1) — RF Analog Tuning ⚙ (Undocumented)	26
0x41 (Page 1) — VCO / PA Control ⚙	27
0x42 (Page 1) — Carrier-Wave Tuning ⚙	27
0x43 (Page 1) — OTP-Dependent PA Tuning ⚙	27
0x46 (Page 1) — PA Bias Control ⚙	27
0x47 (Page 1) — OTP-Dependent PA Tuning ⚙	27
0x48 (Page 1) — TX Power Control ⚙	27
0x4C (Page 1) — RF Analog Tuning ⚙	28
0x6D (Page 1) — Calibration Status: Phase 1 Done ⚙	28
0x70 (Page 1) — Calibration Status: VCO Done ⚙	28
0x7F (Page 1) — Calibration Status: Frequency / Phase 2 Done ⚙	28

Initialization Sequence	28
TX Operation	30
RX Operation	30
TX Power Configuration	30
Carrier-Wave Test Mode	31
Sleep / Wakeup (09sleep example)	31
BLE Mode Configuration Differences	31

PAN211x Register Reference

Sources: official reference manual, PAN211x-DK-v2.2.5 SDK (`pan211.c` and related examples).
 Notation: ⚙️ = SDK-derived / inferred from code; entries with no qualifier are from the official RM.

Address Space and Page Concept

The chip has a 7-bit register address space (0x00–0x7F, 128 locations). Two independent banks exist behind those addresses:

Page	Selected by	Contents
Page 0 (default)	PAGE_CFG = 0x00	User configuration and status registers
Page 1	PAGE_CFG = 0x01	Analog RF calibration and TX-power trim

Two registers are **shared** — they ignore PAGE_CFG and are always accessible: - PAGE_CFG (0x00) - STATE_CFG (0x02)

I2C Access Byte Encoding

The PAN211x I2C address is **0x71** (7-bit). The first data byte in every I2C transaction is a register **access byte**, not a plain register address:

```
access_byte = reg_addr << 1 | R/W    (0=write, 1=read)
```

Write to register 0x07: START 0xE2 0x0E <data> STOP

Read from register 0x73: START 0xE2 0xE6 RESTART 0xE3 <data> STOP

Page 0 Register Map

Addr	Name	R/W	Default	Summary
0x00	PAGE_CFG	R/W	0x00	Page bank select (shared)
0x01	TRX_FIFO	R/W	0x00	TX/RX FIFO access point
0x02	STATE_CFG	R/W	0x00	State machine control (shared)
0x03	SYS_CFG	R/W	0x02	System control / soft reset
0x04	SPI_CFG	R/W	0x6B	SPI interface config, pull-ups
0x05	XTAL_CFG	R/W	–	Crystal load-cap trim
0x06	I2C_CFG	R/W	0x05	I2C interface config
0x07	WMODE_CFG0	R/W	0x49	CRC mode, protocol, whitening, endian
0x08	WMODE_CFG1	R/W	0x83	FIFO size, DPL, enhanced, address width
0x09	RXPllen_CFG	R/W	0x00	Fixed RX payload length
0x0A	TXPllen_CFG	R/W	0x00	Fixed TX payload length
0x0B	RFIRQ_CFG	R/W	0x00	Interrupt mask
0x0C	PID_CFG	R/W	0x00	PID manual control, address-error threshold
0x0D	TRXTWTL_CFG	R/W	0x00	TX→RX switch wait time [7:0]
0x0E	TRXTWTH_CFG	R/W	0x00	TX→RX switch wait time [14:8]
0x0F	PIPE0_RXADDR0	R/W	0xCC	Pipe 0 RX address byte 0 (LSB, first on air)
0x10	PIPE0_RXADDR1	R/W	0xCC	Pipe 0 RX address byte 1
0x11	PIPE0_RXADDR2	R/W	0xCC	Pipe 0 RX address byte 2
0x12	PIPE0_RXADDR3	R/W	0xCC	Pipe 0 RX address byte 3
0x13	PIPE0_RXADDR4	R/W	0xCC	Pipe 0 RX address byte 4 (MSB)
0x14	TXADDR0	R/W	0xCC	TX destination address byte 0 (LSB)
0x15	TXADDR1	R/W	0xCC	TX destination address byte 1
0x16	TXADDR2	R/W	0xCC	TX destination address byte 2
0x17	TXADDR3	R/W	0xCC	TX destination address byte 3
0x18	TXADDR4	R/W	0xCC	TX destination address byte 4 (MSB)
0x19	PKT_EXT_CFG	R/W	0x00	Auto-insert header, FEC / spread-spectrum
0x1A	WHITEN_CFG	R/W	0x7F	Whitening LFSR seed and skip-address flag
0x1B	TXHDR0_CFG	R/W	0x00	Auto-inserted TX header byte 0
0x1C	TXHDR1_CFG	R/W	0x00	Auto-inserted TX header byte 1
0x1D	TXRAMADDR_CFG	R/W	0x00	TX FIFO RAM start address
0x1E	RXRAMADDR_CFG	R/W	0x00	RX FIFO RAM start address
0x1F	RXPIPE_CFG	R/W	0x01	Multi-pipe RX enable
0x20	PIPE1_RXADDR0	R/W	0xCC	Pipe 1 address byte 0 (LSB)
0x21	PIPE1_RXADDR1	R/W	0xCC	Pipe 1 address byte 1
0x22	PIPE1_RXADDR2	R/W	0xCC	Pipe 1 address byte 2
0x23	PIPE1_RXADDR3	R/W	0xCC	Pipe 1 address byte 3
0x24	PIPE1_RXADDR4	R/W	0xCC	Pipe 1 address byte 4 (MSB)
0x25	PIPE2_RXADDR0	R/W	0xCC	Pipe 2 address byte 0 (MSBs shared with Pipe 1)
0x26	PIPE3_RXADDR0	R/W	0xCC	Pipe 3 address byte 0

Addr	Name	R/W	Default	Summary
0x27	PIPE4_RXADDR0	R/W	0xCC	Pipe 4 address byte 0
0x28	PIPE5_RXADDR0	R/W	0xCC	Pipe 5 address byte 0
0x29	TXAUTO_CFG	R/W	0x03	Auto-retransmit delay and count
0x2A	TRXMODE_CFG	R/W	0x01	TX/RX mode, pre-sync options
0x2B	RXTIMEOUTL_CFG	R/W	0xD0	RX timeout [7:0] in μ s
0x2C	RXTIMEOUTH_CFG	R/W	0x07	RX timeout [15:8] in μ s
0x2D	BLEMATCH_CFG0	R/W	0x00	BLE sniffer, whitelist filter, length filter
0x2E	BLEMATCH_CFG1	R/W	0x28	Reserved – do not modify
0x2F	WLIST0_CFG	R/W	0x00	BLE whitelist AdvA byte 0
0x30	WLIST1_CFG	R/W	0x00	BLE whitelist AdvA byte 1
0x31	WLIST2_CFG	R/W	0x00	BLE whitelist AdvA byte 2
0x32	WLIST3_CFG	R/W	0x00	BLE whitelist AdvA byte 3
0x33	WLIST4_CFG	R/W	0x00	BLE whitelist AdvA byte 4
0x34	WLIST5_CFG	R/W	0x00	BLE whitelist AdvA byte 5
0x35	BLEMATCHSTART_CFG	R/W	0x07	BLE whitelist filter payload start offset
0x36	RF_DATARATE_CFG	R/W	0x55	Air data rate
0x37–0x38	–	–	–	Undocumented; not written by SDK
0x39	RF_CHANNEL_CFG	R/W	0x00	RF channel: $F = 2400 + \text{val}$ [MHz]
0x3A–0x42	⊗	W	–	RF analog tuning – undocumented
0x43	⊗	W	–	RF driver tuning; TX-power dependent
0x44	⊗	W	–	RF output tuning; TX-power dependent
0x45	IRQ_MUX_CFG	R/W	0x00	IRQ pin: interrupt / clock-out / PA control
0x46–0x6E	⊗	W	–	RF analog tuning – undocumented
0x6F	MISC_CFG	R/W	0x00	ACK pipe number, IRQ polarity
0x70–0x72	–	–	–	Undocumented
0x73	RFIRQFLG	R/W	0x00	Interrupt flags (write 1 to clear)
0x74	STATUS0	R	0x0C	RX pipe number, PID
0x75	STATUS1	R	0x00	Received header byte 0
0x76	STATUS2	R	0x00	Received header byte 1
0x77	STATUS3	R	0x00	Received payload length
0x78–0x79	–	–	–	Undocumented
0x7A	PKT_RSSI_L	R	0x00	Last-packet RSSI [7:0]
0x7B	PKT_RSSI_H	R	0x00	Last-packet RSSI [13:8]
0x7C–0x7D	–	–	–	Undocumented
0x7E	RT_RSSI_L	R	0x00	Ambient noise RSSI [7:0]
0x7F	RT_RSSI_H	R	0x00	Ambient noise RSSI [13:8]

Page 0 Register Details

0x00 — PAGE_CFG — Page Bank Select

Shared register — accessible from both pages.

Bits	Name	R/W	Default	Description
7:1	—	—	0	Reserved
0	PAGE_SEL	R/W	0	0 = Page 0 (user registers); 1 = Page 1 (analog RF)

Always restore to 0x00 after any Page 1 access.

0x01 — TRX_FIFO — TX/RX FIFO Access

Bits	Name	R/W	Default	Description
7:0	FIFO_DATA	R/W	0x00	Burst-write: loads TX FIFO. Burst-read: drains RX FIFO.

The chip maintains separate TX and RX FIFO pointers. In the I2C implementation, each byte is a separate START-STOP transaction; the chip advances its internal pointer on each transaction. The active FIFO (TX or RX) is selected by the current operating mode.

TX: write `len` bytes before asserting TX mode. Data replaces whatever was in the FIFO.

RX: read `len` bytes after `RX_IRQ` fires.

0x02 — STATE_CFG — State Machine Control

Shared register — accessible from both pages.

Bits	Name	R/W	Default	Description
7	TX_FIFO_READY	R/W	0	✱ Write 1 to trigger TX in continuous TX mode (TX_CFG_MODE=1). Self-clearing.
6	EN_LS_3V	R/W	0	✱ High-voltage module enable.
5	POR_RSTL	R/W	0	✱ Low-voltage domain reset (active low).
4	ISO_TO_0	R/W	0	✱ Register signal isolation enable.
3	—	—	0	Reserved
2:0	OPERATE_MODE	R/W	0	Operating state – see table below

OPERATE_MODE values:

Value	State	Description
0	Deep Sleep	Lowest power; all clocks off; register contents lost
1	Sleep	Low power; register contents retained
2	STB1	Standby 1
3	STB2	Standby 2
4	STB3	Standby 3 – primary idle state; registers accessible
5	TX	Transmit
6	RX	Receive (continuous or single, per TRXMODE_CFG)

Known write values:

Value	Meaning
0x04	Initial STB3 entry (bits 2:0 = 4, high bits = 0) – used at start of Init
0x74	STB3 with EN_LS_3V=1 (bit 6) – normal idle after init
0x75	TX mode with EN_LS_3V=1
0x76	RX mode with EN_LS_3V=1
0x21	Enter sleep (PAN211_EnterSleep)
0x22	Exit sleep / wake-up (PAN211_ExitSleep)

Always write STB3 before reconfiguring registers. The chip must not be in TX or RX when address, payload-length, or mode registers are changed.

0x03 – SYS_CFG – System Control

Bits	Name	R/W	Default	Description
7:3	–	–	–	Reserved – do not modify
2	IRQ_DATA_MUX_EN	R/W	0	Multiplex DATA pin with IRQ output
1	SOFT_RSTL	R/W	1	Logic soft-reset (active low). Write 0 to assert, write 1 to release.
0	–	–	0	Reserved

Init sequence:

Write 0x00 (assert reset) → 1 ms delay → Write 0x02 (release reset) → Write 0x06 (IRQ_DATA_MUX_EN=1 plus release reset).

After soft reset all Page 0 registers return to defaults. SPI_CFG must be re-written.

0x04 — SPI_CFG — SPI / Bus Interface Config

Bits	Name	R/W	Default	Description
7	REG_SPI3_REN	R/W	0	Enable 3-wire SPI read mode. Must be set before entering STB3. In 4-wire SPI the DATA line is bidirectional; in 3-wire mode separate MOSI/MISO are used. Not meaningful for I2C operation but the SDK always sets this bit (0x83).
6	REG_DATA_PUEN	R/W	1	DATA/SDA pull-up enable
5	REG_CSN_PUEN	R/W	1	CSN pull-up enable
4	REG_SCK_PUEN	R/W	1	SCK/SCL pull-up enable
3	REG_IN_PAD_MODE	R/W	0	Manual pad configuration mode
2:0	—	—	011	Reserved – forbidden to modify; always include 0b011

Default = 0x6B = REG_DATA_PUEN | REG_CSN_PUEN | REG_SCK_PUEN | 0b011.

SDK init writes **0x83** = REG_SPI3_REN | 0b011. After soft-reset this register reads back 0x83; the firmware uses this as a chip-present check.

Page 1 dual use: When PAGE_CFG = 1, address 0x04 maps to the OTP data register (see Page 1 section).

0x05 — XTAL_CFG — Crystal Load-Capacitor Trim

Bits	Name	R/W	Default	Description
7:6	—	⊗	0b11	⊗ SDK always ORs 0xC0 into this register – these bits appear to be a required constant or enable field.
5:0	XTAL_TRIM	R/W	—	Crystal load-capacitor trim value. Loaded from OTP: (value4 >> 4) 0xC0 where value4 is the factory-programmed OTP word.

Write = (OTP_value4 >> 4) | 0xC0. The upper 4 bits of OTP word 4 carry the factory-calibrated trim. Operating with the wrong trim degrades frequency accuracy and may prevent BLE-compatible reception.

Page 1 dual use: When PAGE_CFG = 1, address 0x05 maps to the OTP mode control register (see Page 1 section).

0x06 — I2C_CFG — I2C Interface Config

Bits	Name	R/W	Default	Description
7:4	—	—	0	Reserved
3	IRQ_I2C_MUX_EN	R/W	0	Multiplex IRQ signal onto the I2C SDA pin
2:0	—	—	0b101	Reserved – forbidden to modify

Default = 0x05 (reserved bits 2:0 = 0b101). SDK does not write this register; the default is sufficient for I2C operation.

0x07 — WMODE_CFG0 — Work Mode Configuration 0

Bits	Name	R/W	Default	Description
7:6	CRC_MODE	R/W	01	CRC configuration: 00 =off, 01 =1 byte, 10 =2 bytes, 11 =3 bytes (BLE)
5:4	WORK_MODE	R/W	00	Protocol: 00 =XN297L-compatible, 11 =BLE. Values 01 , 10 not defined.
3	WHITEN_EN	R/W	1	Enable data whitening using LFSR seed from WHITEN_CFG
2	CRC_SKIP_ADDR	R/W	0	Exclude address field from CRC calculation
1	TX_NOACK	R/W	0	Do not wait for ACK in enhanced mode
0	ENDIAN	R/W	1	Bit order: 0 =little-endian (BLE), 1 =big-endian (XN297L-compatible)

Mode values used in practice:

Value	Binary	CRC	WORK_MODE	Whiten	Endian	Use
0x81	10000001	2B	XN297L	off	big	XN297L, no whitening
0x89	10001001	2B	XN297L	on	big	SDK normal TX (00_normal_tx)
0xFC	11111100	3B	BLE	on	little	SDK BLE mode

0x08 — WMODE_CFG1 — Work Mode Configuration 1

Bits	Name	R/W	Default	Description
7	RX_G00N	R/W	1	Stay in RX after packet received (or after CRC/address error). If 0, chip returns to STB3 after each received packet.
6	PRI_EXIT_RX	R/W	0	Force exit from RX immediately (write 1 to abort RX)
5	FIFO_128_EN	R/W	0	1 = 128-byte FIFO; 0 = 64-byte FIFO
4	DPY_EN	R/W	0	Dynamic payload length: 1 = auto-extract length from received PDU header; 0 = use RXPLEN_CFG fixed length
3	ENHANCE	R/W	0	Enhanced mode: auto-ACK, PID duplicate detection, max-retransmit
2	—	—	0	Reserved
1:0	ADDR_BYTE_LEN	R/W	11	Address width: 00 =2 B, 01 =3 B, 10 =4 B, 11 =5 B

Mode values used in practice:

Value	Binary	RX_GOON	FIFO	DPY	ENHANCE	Addr	Use
0xA3	10100011	1	128B	0	0	5B	SDK normal TX (00_normal_tx)
0xB2	10110010	1	128B	1	0	4B	SDK BLE mode
0x9B	10011011	1	64B	1	1	5B	SDK enhanced mode
0x83	10000011	1	64B	0	0	5B	Power-on default

0x09 — RXPLEN_CFG — Receive Payload Length

Bits	Name	R/W	Default	Description
7:0	RX_PAYLOAD_LEN	R/W	0x00	Fixed receive payload length in bytes. Ignored when DPY_EN =1.

Range: 0-128 (when FIFO_128_EN=1) or 0-64.

0x0A — TXPLEN_CFG — Transmit Payload Length

Bits	Name	R/W	Default	Description
7:0	TX_PAYLOAD_LEN	R/W	0x00	Number of bytes to transmit from FIFO. Must be written before every TX if length varies.

0x0B — RFIRQ_CFG — Interrupt Mask

All bits: 0 = interrupt **enabled** (asserts IRQ pin); 1 = interrupt **masked** (suppressed).

Bits	Name	R/W	Default	Description
7	TX_IRQ_MSK	R/W	0	Mask TX-complete interrupt
6	TX_MAX_RT_IRQ_MSK	R/W	0	Mask max-retransmit interrupt (enhanced mode)
5	RX_ADDR_ERR_MSK	R/W	0	Mask address-mismatch interrupt (FEC/spread-spectrum mode)
4	RX_CRC_ERR_IRQ_MSK	R/W	0	Mask CRC-error interrupt
3	RX_LEN_ERR_IRQ_MSK	R/W	0	Mask length-error interrupt (enhanced mode)
2	RX_PID_ERR_IRQ_MSK	R/W	0	Mask duplicate-PID interrupt (enhanced mode)
1	RX_TIMEOUT_IRQ_MSK	R/W	0	Mask RX-timeout interrupt
0	RX_IRQ_MSK	R/W	0	Mask RX-complete interrupt

Common init values:

Value	TX	RX	Errors	Use
0x7F	masked	enabled	masked	SDK normal TX (00_normal_tx)
0xEE	enabled	enabled	masked	SDK realtime RSSI (19_realtime_rssi)
0x6C	enabled	enabled	partial	SDK sleep example (09_sleep)
0x28	masked	masked	partial	SDK enhanced-mode RX timeout only

0x0C — PID_CFG — PID Identifier Configuration

Bits	Name	R/W	Default	Description
7	PID_MANUAL_EN	R/W	0	Enable manual PID control (override auto-increment)
6:4	ADDR_ERR_THR	R/W	0	Address-match error threshold (number of bit errors tolerated)
3:2	RX_PID_MANUAL	R/W	0	PID value to expect in received packets (when PID_MANUAL_EN=1)
1:0	TX_PID_MANUAL	R/W	0	PID value to embed in transmitted packets (when PID_MANUAL_EN=1)

PID (packet ID) is used in enhanced mode to detect duplicate packets. In normal and BLE modes this register is left at 0x00.

0x0D — TRXTWTL_CFG — TX/RX Switch Wait Time (Low)

0x0E — TRXTWTH_CFG — TX/RX Switch Wait Time (High)

Reg	Bits	Name	R/W	Default	Description
0x0D	7:0	TRX_WAIT[7:0]	R/W	0x00	Low byte of 15-bit wait-time field
0x0E	6:0	TRX_WAIT[14:8]	R/W	0x00	High 7 bits; bit 7 reserved

Specifies a wait period (in μ s) between TX→RX or RX→TX transitions. Default = 0.

0x0F-0x13 — PIPE0_RXADDR0-4 — Pipe 0 RX Address

Five-byte hardware address filter for Pipe 0. Only packets whose sync-word matches this address are forwarded to the FIFO and assert `RX_IRQ`.

Byte 0 (0x0F) is the LSB and the first byte sent over the air. Byte 4 (0x13) is the MSB. When using 4-byte addresses (`ADDR_BYTE_LEN = 10`), byte 4 is ignored; when using 2-byte addresses only bytes 0-1 are used.

Default = 0xCCCCCCCCC.

The address doubles as the RF sync word — a receiving device only accepts packets where the sender's TXADDR matches its PIPE0_RXADDR.

0x14-0x18 — TXADDR0-4 — TX Destination Address

Destination address embedded in the packet as the sync word. Must match the intended receiver's `PIPE0_RXADDR` for hardware filtering to pass. Same byte order as `PIPE0_RXADDR`. Updated before each TX when sending to different destinations.

Default = 0xCCCCCCCCC.

0x19 — PKT_EXT_CFG — Packet Extension Configuration

Bits	Name	R/W	Default	Description
7	W_RX_MAX_CTRL_EN	R/W	0	✱ Enable extended RX length control
6	HDR_LEN_EXIST	R/W	0	Auto-insert header bytes from TXHDR0/1 into every TX packet. When 1, FIFO must contain only payload (no header prefix).
5:4	HDR_LEN_NUMB	R/W	00	Number of header bytes to auto-insert: 00 =0, 01 =1, 10 =2
3	PRI_TX_FEC	R/W	0	TX spread-spectrum / FEC enable
2	PRI_RX_FEC	R/W	0	RX spread-spectrum / FEC enable
1:0	PRI_CI_MODE	R/W	00	Spread-spectrum mode: 00 =off, 01 =S2, 10 =S8

BLE TX mode: `PKT_EXT_CFG = 0x60 = HDR_LEN_EXIST=1, HDR_LEN_NUMB=10` (2 header bytes).

The chip auto-prepends `TXHDR0` (PDU type = `0x42 = ADV_NONCONN_IND | TxAdd=1`) and `TXHDR1` (auto-calculated PDU length) to the FIFO payload. The FIFO must contain only AdvA (6 bytes, LSB-first) + AdvData.

0x1A — WHITEN_CFG — Whitening Configuration

Bits	Name	R/W	Default	Description
7	WHITEN_SKIP_ADDR	R/W	0	1 = whitening skips the address field and starts at the payload. Required for BLE (preamble and access address are never whitened in BLE).
6:0	WHITEN_SEED	R/W	0x7F	Whitening LFSR initial value. Both TX and RX must use the same seed.

Known values:

Value	SKIP_ADDR	SEED	Use
0x7F	0	0x7F	SDK default / XN297L-compatible
0x53	0	0x53	BLE advertising channel 37 (2402 MHz)
0x33	0	0x33	BLE advertising channel 38 (2426 MHz)
0x73	0	0x73	BLE advertising channel 39 (2480 MHz)

BLE whitening seed formula:

```
WHITEN_CFG = bit_reverse7(BLE_channel_index | 0x40)
```

where `bit_reverse7` reverses all 7 bits and `BLE_channel_index` is the BLE logical channel number (37, 38, or 39 for advertising). In BLE `WORK_MODE` (`WORK_MODE = 11`) address-field skipping is handled automatically by the chip; `WHITEN_SKIP_ADDR` (bit 7) is left 0.

Example for BLE ch 37: $37 \mid 0x40 = 101 = 0b1100101$; bit-reversed (7 bits) = $0b1010011 = \mathbf{0x53}$.

0x1B — TXHDR0_CFG — TX Header Byte 0

Bits	Name	R/W	Default	Description
7:0	TX_HEADER0	R/W	0x00	Value auto-inserted as the first header byte when <code>PKT_EXT_CFG.HDR_LEN_EXIST = 1</code>

BLE advertising: `TXHDR0_CFG = 0x42 = ADV_NONCONN_IND (PDU type 0x02) | TxAdd=1 (bit 6)`.
(`0b01000010` → bits [5:4]=`0b10` reserved, bits[3:0]=PDU type `0x2=ADV_NONCONN_IND`, bit[6]=`TxAdd=1`)

Note: On Page 1, address 0x1B maps to the **Calibration Control** register (different physical register — see Page 1 section).

0x1C — TXHDR1_CFG — TX Header Byte 1

Bits	Name	R/W	Default	Description
7:0	TX_HEADER1	R/W	0x00	Value auto-inserted as the second header byte when <code>HDR_LEN_NUMB ≥ 2</code> . In BLE mode the chip fills this with the actual PDU payload length automatically.

0x1D — TXRAMADDR_CFG — TX FIFO RAM Start Address

Bits	Name	R/W	Default	Description
7:0	TX_RAM_START	R/W	0x00	Byte offset into the shared FIFO RAM where the TX payload begins. Normally 0x00.

0x1E — RXRAMADDR_CFG — RX FIFO RAM Start Address

Bits	Name	R/W	Default	Description
7:0	RX_RAM_START	R/W	0x00	Byte offset into the shared FIFO RAM where received data is placed. Normally 0x00.

0x1F — RXPIPE_CFG — Multi-Pipe RX Enable

Bits	Name	R/W	Default	Description
7:6	—	—	—	Reserved — forbidden
5	PIPE5_EN	R/W	0	Enable RX pipe 5
4	PIPE4_EN	R/W	0	Enable RX pipe 4
3	PIPE3_EN	R/W	0	Enable RX pipe 3
2	PIPE2_EN	R/W	0	Enable RX pipe 2
1	PIPE1_EN	R/W	0	Enable RX pipe 1
0	PIPE0_EN	R/W	1	Enable RX pipe 0 (always-on default)

Write 0x01 to enable Pipe 0 only (typical single-address use). Multi-pipe mode allows simultaneous reception on up to 6 different addresses.

Pipes 2–5 share MSBs with Pipe 1 (only their LSB differs — PIPE2_RXADDR0 – PIPE5_RXADDR0).

0x20–0x24 — PIPE1_RXADDR0–4 — Pipe 1 RX Address

Full 5-byte address for Pipe 1. Same byte order as PIPE0_RXADDR. Default = 0xCCCCCCCCC.

0x25–0x28 — PIPE2–5_RXADDR0 — Pipes 2–5 Address LSB

Pipes 2–5 each have only their LSB (byte 0) independently programmable; bytes 1–4 are copied from Pipe 1's address.

Reg	Pipe
0x25	PIPE2_RXADDR0
0x26	PIPE3_RXADDR0
0x27	PIPE4_RXADDR0
0x28	PIPE5_RXADDR0

0x29 — TXAUTO_CFG — Auto-Retransmit Configuration

Bits	Name	R/W	Default	Description
7:4	ARD	R/W	0000	Auto-retransmit delay: $250\ \mu\text{s} \times (\text{ARD} + 1)$. Range: $250\ \mu\text{s}$ – $4000\ \mu\text{s}$.
3:0	ARC	R/W	0011	Auto-retransmit count: number of additional TX attempts (0 = no retransmit, 14 = max). Value 15 is reserved.

Only used in enhanced mode (ENHANCE =1). Normal and BLE modes write 0x00 (no retransmit).

0x2A — TRXMODE_CFG — TX/RX Mode Configuration

Bits	Name	R/W	Default	Description
7	REG_TX_CFG_MODE	R/W	0	TX mode: 0 =single burst (chip returns to STB3 after packet); 1 =continuous carrier
6:5	REG_RX_CFG_MODE	R/W	00	RX mode (normal): 00 =single; 01 =single+timeout; 10 =continuous RX mode (enhanced): 00 =continuous; 01 =continuous+timeout
4	PRE_2BYTE_MODE	R/W	0	✱ Use 2-byte preamble instead of 1-byte
3	W_PRE_SYNC_12B_EN	R/W	0	Enable 12-bit pre-sync pattern
2	W_PRE_SYNC_8B_EN	R/W	0	Enable 8-bit pre-sync pattern
1	W_PRE_SYNC_4B_EN	R/W	0	Enable 4-bit pre-sync pattern
0	W_PRE_SYNC_EN	R/W	1	Enable pre-sync (preamble detect). Default=1; must remain 1 for normal operation.

Common values:

Value	TX	RX	Use
0x41	single	continuous	SDK normal mode
0x81	continuous	—	Carrier-wave test mode
0x61	single	continuous+timeout	SDK enhanced mode

0x2B — RXTIMEOUTL_CFG — RX Timeout (Low)**0x2C — RXTIMEOUTH_CFG — RX Timeout (High)**

16-bit timeout value in microseconds. RX_TIMEOUT_IRQ fires if no packet is received within this window (requires REG_RX_CFG_MODE to include timeout).

Reg	Bits	Default
0x2B	[7:0]	0xD0
0x2C	[15:8]	0x07

Default = 0x07D0 = 2000 μ s (2 ms). Write low byte first.

0x2D — BLEMATCH_CFG0 — BLE Filter Configuration

Bits	Name	R/W	Default	Description
7	SNIF_EN	R/W	0	Sniffer mode: accept all packets regardless of address
6:4	WL_MATCH_MODE	R/W	000	Whitelist filter depth: 000 =disabled; 001 =compare bits[47:40]; 010 =bits[47:32]; 011 =bits[47:24]; 100 =bits[47:16]; 101 =bits[47:8]; 110 =full 48 bits
3:2	BLELEN_MATCH_MODE	R/W	00	Length filter: 00 =disabled; 01 =equal; 10 =greater-than; 11 =less-than
1:0	—	—	00	Reserved

Filter compares the received AdvA field against WLST0-5_CFG . Only used in BLE mode (WORK_MODE =11).

SDK BLE TX init: 0x04 (length filter = equal, whitelist disabled).

0x2E — BLEMATCH_CFG1

Bits	Default	Note
7:0	0x28	Reserved – do not modify. Always leave at 0x28.

0x2F-0x34 — WLST0-5_CFG — BLE Whitelist AdvA

Six-byte BLE advertiser address (AdvA) used by the hardware whitelist filter. Byte 0 (0x2F) = bits[7:0], byte 5 (0x34) = bits[47:40].

Only used in BLE mode with WL_MATCH_MODE ≠ 000.

0x35 — BLEMATCHSTART_CFG — BLE Filter Payload Start

Bits	Name	R/W	Default	Description
7:6	—	—	—	Reserved – forbidden
5:0	PLD_START_BYTE	R/W	0x07	Byte offset into the received packet where the whitelist comparison begins. Default = 7.

SDK BLE TX init writes 0x00 (start from byte 0).

0x36 — RF_DATARATE_CFG — Air Data Rate

Bits	Name	R/W	Default	Description
7:6	—	—	01	Reserved constant – always 01 (include in write)
5:4	DATARATE	R/W	00	Air data rate: 00 =1 Mbps, 01 =2 Mbps, 11 =250 kbps
3:0	—	—	0101	Reserved constant – always 0101 (include in write)

Full register constants (including reserved bits):

Constant	Value	Rate
DataRate1Mbps	0x55	1 Mbps
DataRate2Mbps	0x65	2 Mbps
DataRate250kbps	0x75	250 kbps

BLE requires 1 Mbps.

0x37-0x38 — Undocumented

Not described in the RM. Purpose inferred from hardware testing.

0x37 — Crystal pre-configuration gate (16 MHz hardware)

Must be written to **0xE0** before entering Page 1 for OTP access when using a 16 MHz crystal. Without this write, the Page 1 RF analog tuning values and OTP read do not take effect correctly. Not required (and not written) by the SDK, which targets 32 MHz crystal hardware.

0x38 — Purpose unknown; not written by SDK or 16 MHz init.

0x39 — RF_CHANNEL_CFG — RF Channel

Bits	Name	R/W	Default	Description
7:0	RF_CH	R/W	0x00	Channel number. Center frequency = 2400 + RF_CH [MHz].

Valid range: 0–83 (2400–2483 MHz, covering the full 2.4 GHz ISM band).

BLE advertising channels:

BLE Channel	RF_CH	Frequency
37	0x02	2402 MHz
38	0x1A	2426 MHz
39	0x50	2480 MHz

Calibration channel: 0x55 = 2485 MHz (outside operating range; used during RF calibration to avoid interference).

0x3A-0x42 — RF Analog Tuning (Undocumented, Page 0)

⚙️ These registers are not documented in the official RM. The SDK writes them during initialization with values derived from the ES_Tool V1.2.6 for a 16 MHz crystal. Do not modify without guidance from PANCHIP.

Not written by the SDK init sequence (the range 0x3A–0x42 is skipped; SDK jumps directly from 0x39 to 0x43).

0x43 — RF Driver Tuning (Undocumented, Page 0) ⚙

SDK Init	TX 0 dBm	TX 9 dBm	Notes
0x3A	0x3A	0x3A	Same value for all TX-power levels; from ES_Tool init sequence

Inferred purpose: RF driver bias or gain setting. Changing this register requires corresponding Page 1 register adjustments.

0x44 — RF Output Tuning (Undocumented, Page 0) ⚙

SDK Init	TX 0 dBm	TX 9 dBm	Notes
0x8C	0x84	0x8C	TX-power dependent – part of TX-power configuration

Inferred purpose: PA output matching or bias. The difference between 0 dBm (0x84) and 9 dBm (0x8C) confirms a direct role in output power control.

0x45 — IRQ_MUX_CFG — IRQ Pin Multiplexing

Bits	Name	R/W	Default	Description
7:4	–	–	–	Reserved – forbidden
3:2	OCLK_SEL	R/W	00	Clock frequency when IRQ_MUX = 01: 00 = 1 kHz, 01 = 4 kHz, 10 = 8 MHz, 11 = 16 MHz
1:0	IRQ_MUX	R/W	00	IRQ pin function: 00 = interrupt output, 01 = clock output, 10 = PA control signal

Not required when IRQ is polled via register read rather than used as a hardware interrupt.

0x46-0x54 — RF Analog Tuning (Undocumented, Page 0) ⚙

Not written by the SDK in the ES_Tool init sequence for normal operation.

0x55-0x61 — RF Analog Tuning Bank (Undocumented, Page 0) ⚙

Written during init from the SDK ES_Tool V1.2.6 sequence (16 MHz crystal). These are required for correct RF operation and must not be changed.

Addr	SDK Init	BLE RX mode	Inferred function
0x55	0xDD	0xDD	RF filter / baseband tuning
0x56	0xC9	0xC9	RF filter / baseband tuning
0x57	0xB7	0xB7	RF filter / baseband tuning
0x58	—	—	Not written; purpose unknown
0x59	—	—	Not written; purpose unknown
0x5A	0x10	0x10	RF baseband tuning
0x5B	0xFD	0xFD	RF baseband tuning
0x5C	0xE9	0xE9	RF baseband tuning
0x5D	0xDC	0xD4 ☸	RF baseband tuning; 0xD4 used in 19_realtime_rssi (high-gain RX)
0x5E	0x02	0x02	RF baseband tuning
0x5F	0x06	0x06	RF baseband tuning
0x60	0x0E	0x0E	RF baseband tuning
0x61	0x2E	0x3E ☸	RF baseband / RX gain: 0x3E = high-gain RX mode (used in 19_realtime_rssi); 0x2E = normal

0x62-0x65 — RF Analog Tuning (Undocumented, Page 0) ☸

Not written by SDK. Purpose unknown.

0x66 — RF Analog Tuning (Undocumented, Page 0) ☸

SDK init value: **0x34**. Purpose unknown; required for correct RF operation.

0x67 — RF Analog Tuning (Undocumented, Page 0) ☸

Not written by SDK. Purpose unknown.

0x68 — RF Analog Tuning (Undocumented, Page 0) ☸

SDK init value: **0x0D**. Purpose unknown; required for correct RF operation.

0x69-0x6D — RF Analog Tuning (Undocumented, Page 0) ☸

Not written by SDK. Purpose unknown.

0x6E — RF Analog Tuning (Undocumented, Page 0) ☸

SDK init value: **0x20**. Last register in the ES_Tool init sequence before the Page 1 RF calibration step.

0x6F — MISC_CFG — Miscellaneous Configuration

Bits	Name	R/W	Default	Description
7	—	—	0	Reserved
6	ENH_NOACK_RX_CONT_DIS	R/W	0	✖ Disable continuous RX after no-ACK in enhanced mode
5	I_NDC_PREAMBLE_SEL	R/W	0	✖ Preamble type select
4	PID_LOW_SEL	R/W	0	✖ PID comparison mode
3	IRQ_HIGH_EN	R/W	0	IRQ pin polarity: 0=active low (default), 1=active high
2:0	ACK_PIPE	R/W	0	Pipe number to use for ACK packets in enhanced mode

SDK BLE RX example: writes **0x10** to this register — PID_LOW_SEL =1.

0x70-0x72 — Undocumented (Page 0)

Not described in RM. Not written during normal operation. On Page 1, address 0x70 is the VCO calibration status register (see Page 1 section).

0x73 — RFIRQFLG — RF Interrupt Status Flags

Write 1 to any bit to clear that flag. Flags are set by hardware and cleared by firmware.

Bits	Name	R/W	Default	Description
7	TX_IRQ	R/W	0	TX complete: packet sent successfully. In single TX mode, chip has returned to STB3.
6	TX_MAX_RT_IRQ	R/W	0	Max retransmits reached (enhanced mode) – TX failed; chip in STB3
5	RX_ADDR_ERR_IRQ	R/W	0	Address match error in FEC/spread-spectrum mode
4	RX_CRC_ERR_IRQ	R/W	0	CRC check failed on received packet
3	RX_LEN_ERR_IRQ	R/W	0	Payload length error (enhanced mode)
2	RX_PID_ERR_IRQ	R/W	0	Duplicate PID received (enhanced mode)
1	RX_TIMEOUT_IRQ	R/W	0	RX timeout expired without a packet
0	RX_IRQ	R/W	0	Valid packet received and available in FIFO

Write **0xFF** to clear all flags. Write **0x80** to clear TX_IRQ only. Write **0x01** to clear RX_IRQ only.

SDK-defined constants (pan211.h):

Constant	Value	Flag
RF_IT_TX_IRQ	0x80	TX complete
RF_IT_MAX_RT_IRQ	0x40	Max retransmit
RF_IT_ADDR_ERR_IRQ	0x20	Address error
RF_IT_CRC_ERR_IRQ	0x10	CRC error
RF_IT_LEN_ERR_IRQ	0x08	Length error
RF_IT_PID_ERR_IRQ	0x04	PID error
RF_IT_RX_TIMEOUT_IRQ	0x02	RX timeout
RF_IT_RX_IRQ	0x01	RX complete
RF_IT_ALL_IRQ	0xFF	All flags

0x74 — STATUS0 — Status Register 0 (Read-Only)

Bits	Name	R	Default	Description
7	RX_CI_ERR	R	0	CI error detected in spread-spectrum mode
6:4	RX_SYNC_ADDR	R	000	Pipe number that received the last packet (0–5); 0x7 = FIFO empty
3:2	RX_PID	R	11	PID value of last received packet
1:0	TX_PID	R	00	PID value of last transmitted packet

Read `RX_SYNC_ADDR` (bits [6:4]) to determine which pipe a packet arrived on in multi-pipe mode.

0x75 — STATUS1 — Received Header Byte 0 (Read-Only)

Bits	R	Default	Description
7:0	R	0x00	First received header byte (from TXHDR0 of the sender). Valid after <code>RX_IRQ</code> . In BLE mode = PDU type byte (e.g., 0x42 = <code>ADV_NONCONN_IND</code>).

0x76 — STATUS2 — Received Header Byte 1 (Read-Only)

Bits	R	Default	Description
7:0	R	0x00	Second received header byte (from TXHDR1 of the sender). In BLE mode = PDU length byte.

0x77 — STATUS3 — Received Payload Length (Read-Only)

Bits	R	Default	Description
7:0	R	0x00	Actual payload byte count stored in FIFO for the last received packet. Read before reading FIFO to know how many bytes to drain. Relevant in dynamic-payload mode.

0x78-0x79 — Undocumented (Page 0)

0x7A — PKT_RSSI_L — Last Packet RSSI (Low)

0x7B — PKT_RSSI_H — Last Packet RSSI (High)

Reg	Bits	R	Description
0x7A	7:0	R	RSSI of last correctly received packet, bits [7:0]
0x7B	5:0	R	RSSI bits [13:8]; bits [7:6] reserved

14-bit RSSI value. Units and reference level not specified in RM. Read after `RX_IRQ` for per-packet signal strength. Larger value = stronger signal.

0x7C-0x7D — Undocumented (Page 0)

0x7E — RT_RSSI_L — Ambient Noise RSSI (Low)

0x7F — RT_RSSI_H — Ambient Noise RSSI (High)

Reg	Bits	R	Description
0x7E	7:0	R	Real-time ambient noise RSSI, bits [7:0]
0x7F	5:0	R	Real-time ambient noise RSSI bits [13:8]; bits [7:6] reserved

Continuous background measurement; readable at any time. Compare with `PKT_RSSI` for SNR estimation.

Collision with Page 1: On Page 1, address 0x7F maps to the **calibration-done status** register (see Page 1 section).

Page 1 Register Details

Page 1 is selected by writing `PAGE_CFG = 0x01`. All entries are  **SDK-derived** — none appear in the official RM.

The same address bits 0x00 and 0x02 remain shared (`PAGE_CFG`, `STATE_CFG`).

0x04 (Page 1) — OTP Data Register

Dual-purpose with Page 0 `SPI_CFG`. On Page 1, used for factory OTP calibration readout.

Write a command byte, then read the result:

Write	Read result
0x04	OTP word 2 (value2)
0x08	OTP word 4 (value4)

OTP readout requires 0x05 (Page 1) to be set to OTP-read mode first.

0x05 (Page 1) — OTP Mode Control ⚙

Value	Meaning
0x00	Enter OTP read mode
0x01	Exit OTP read mode

Sequence: Write 0x05 ← 0x00 → write/read 0x04 → Write 0x05 ← 0x01.

OTP Calibration Data Fields ⚙

OTP word 2 (value2):

Bits	Field	Description
3:0	OTP_VALID	Must equal 0x1 to indicate valid OTP. If not, initialization must abort.
4	CAL_BIT	Used to set bit 0 of Page 1 register 0x43: $\text{calBit} = (\text{value2} \ \& \ 0x10) == 0 ? 1 : 0$
6:5	PA_TRIM	Used to set bits [6:4] of Page 1 register 0x47: $0x83 \mid ((\text{value2} \gg 1) \ \& \ 0x70)$
7	–	Unused

OTP word 4 (value4):

Bits	Field	Description
7:4	XTAL_TRIM	Crystal load-cap trim; written to Page 0 XTAL_CFG : $(\text{value4} \gg 4) \mid 0xC0$
3:0	–	Unused

0x1B (Page 1) — Calibration Control Register ⚙

Shares address with Page 0 TXHDR0_CFG — different physical register.

One-hot write to trigger calibration steps. Poll the corresponding status register until done, then write the next step.

Write	Calibration step	Status register	Done condition	Typical wait
0x08	VCO calibration	0x70 (Page 1)	bit [6] = 1	< 1 ms
0x10	Thermal calibration	–	Fixed delay	55 ms mandatory
0x20	Frequency calibration	0x7F (Page 1)	bit [7] = 1	~1 ms
0x40	Phase calibration 1	0x6D (Page 1)	bit [7] = 1	~2 ms
0x80	Phase calibration 2	0x7F (Page 1)	bit [7] = 1	~2 ms
0x00	Stop calibration	–	–	–

Calibration must run in the exact order above. Frequency calibration requires the chip to be in RX mode (STATE_CFG = 0x76) before writing 0x20. All other steps can run in STB3.

0x27 (Page 1) — RF Analog Tuning ⚙

Init value	TX 0 dBm	TX 9 dBm
0xAA	0xAA	0xAA

Written during Page 1 pre-configuration and during TX-power changes. Constant across power levels.

0x32-0x33 (Page 1) — RF Analog Tuning ⚙

Addr	Value
0x32	0x1E
0x33	0x19

Written during Page 1 pre-configuration. Purpose unspecified; do not modify.

0x37 (Page 1) — RF Analog Tuning ⚙

Init value: **0x15**.

0x3A (Page 1) — RF Analog Tuning ⚙

Init value: **0x14**.

0x3C (Page 1) — TX Power Amplitude Control ⚙

Controls PA output level.

Value	TX power
0x13	0 dBm
0x17	9 dBm

0x3E (Page 1) — RF Analog Tuning ⚙

Init value: **0xF1**.

0x3F (Page 1) — RF Analog Tuning ⚙ (Undocumented)

Not described in the RM or SDK. Required for correct RF operation with a **16 MHz crystal**; init value **0xD2**. Not written by the SDK (which targets 32 MHz hardware).

0x40 (Page 1) — RF Analog Tuning ⚙ (Undocumented)

Not described in the RM or SDK. Required for correct RF operation with a **16 MHz crystal**; init value **0x20**. Not written by the SDK (which targets 32 MHz hardware).

0x41 (Page 1) — VCO / PA Control ⚙

Value	Context
0xA6	Normal operation — 16 MHz crystal
0xA2	Normal operation — 32 MHz crystal (SDK default)
0x20	Carrier-wave test mode entry
0x00	Carrier-wave test mode exit

0x42 (Page 1) — Carrier-Wave Tuning ⚙

Value	Context
0x4E	Carrier-wave test mode
0x00	Normal / exit CW mode

Only written when entering/exiting carrier-wave test mode.

0x43 (Page 1) — OTP-Dependent PA Tuning ⚙

Written from OTP value during initialization:

```
Write(0x43, 0x10 | calBit)
```

where `calBit = (OTP_value2 & 0x10) == 0 ? 1 : 0`.

Possible values: **0x10** (calBit=0) or **0x11** (calBit=1).

0x46 (Page 1) — PA Bias Control ⚙

Value	TX power
0xB0	9 dBm (normal)
0xBD	0 dBm

0x47 (Page 1) — OTP-Dependent PA Tuning ⚙

Written from OTP value during initialization:

```
Write(0x47, 0x83 | ((OTP_value2 >> 1) & 0x70))
```

Base value 0x83; bits [6:4] are filled from OTP `PA_TRIM` field.

0x48 (Page 1) — TX Power Control ⚙

Value	TX power	Notes
0x88	both	Same value for 0 dBm and 9 dBm

0x4C (Page 1) — RF Analog Tuning ⚙

Init value: **0x48**.

0x6D (Page 1) — Calibration Status: Phase 1 Done ⚙

Bit	Meaning
[7]	1 = Phase calibration 1 complete

Poll after writing `CAL_CTL` = 0x40.

0x70 (Page 1) — Calibration Status: VCO Done ⚙

Bit	Meaning
[6]	1 = VCO calibration complete

Poll after writing `CAL_CTL` = 0x08.

0x7F (Page 1) — Calibration Status: Frequency / Phase 2 Done ⚙

Bit	Meaning
[7]	1 = Frequency or Phase 2 calibration complete

Poll after writing `CAL_CTL` = 0x20 (frequency) and again after 0x80 (phase 2).

Collision with Page 0: On Page 0, address 0x7F maps to `RT_RSSI_H`. Always restore `PAGE_CFG` = 0x00 before reading RSSI.

Initialization Sequence

Full sequence from SDK `PAN211_Init()` (`pan211.c`):

Step 1 – Bus init (Page 0)

```
Write PAGE_CFG ← 0x00
Write SPI_CFG ← 0x83 (REG_SPI3_REN=1, required before STB3)
```

Step 2 – STB3 with soft reset

```
Write STATE_CFG ← 0x04 (enter STB3, bits set for low-power enable)
delay 1 ms
Write STATE_CFG ← 0x74
delay 1 ms
Write SYS_CFG ← 0x00 (assert SOFT_RSTL)
delay 1 ms
Write SYS_CFG ← 0x02 (release SOFT_RSTL)
Read SPI_CFG → must read 0x83 (chip-present check)
[16 MHz crystal only] Write 0x37 ← 0xE0 (must precede Page 1 entry)
```

Step 3 – Read factory OTP (Page 1)

```
Write PAGE_CFG ← 0x01
Write 0x05 ← 0x00 (OTP read mode)
Write 0x04 ← 0x04 → Read 0x04 = value2
Write 0x04 ← 0x08 → Read 0x04 = value4
Write 0x05 ← 0x01 (exit OTP mode)
Assert (value2 & 0x0F) == 1 (OTP valid check)
Write 0x47 ← 0x83 | ((value2 >> 1) & 0x70)
Write 0x43 ← 0x10 | ((value2 & 0x10) == 0 ? 1 : 0)
```

Step 4 – Page 1 pre-configuration

```
Write 0x27 ← 0xAA
Write 0x32 ← 0x1E
Write 0x33 ← 0x19
Write 0x37 ← 0x15
Write 0x3A ← 0x14
Write 0x3E ← 0xF1
[16 MHz crystal only] Write 0x3F ← 0xD2
[16 MHz crystal only] Write 0x40 ← 0x20
Write 0x41 ← 0xA6 (16 MHz crystal) / 0xA2 (32 MHz crystal, SDK default)
Write 0x46 ← 0xB0 (= PA_BIAS_9DBM)
Write 0x4C ← 0x48
```

Step 5 – Page 0 RF configuration

```
Write PAGE_CFG ← 0x00
Write XTAL_CFG ← (value4 >> 4) | 0xC0
Write SYS_CFG ← 0x06
Write WMODE_CFG0 ← 0x89 (2B CRC, XN297L, whitening on, big-endian)
Write WMODE_CFG1 ← 0xA3 (RX_G00N, 128B FIFO, 5B addr)
Write RXPLEN_CFG ← PayloadLen
Write TXPLEN_CFG ← PayloadLen
Write RFIRQ_CFG ← 0x7E (TX+RX enabled, errors masked)
Write TXAUTO_CFG ← 0x00 (no retransmit)
Write TRXMODE_CFG ← 0x41 (single TX, continuous RX)
Write WHITEN_CFG ← 0x7F (seed=0x7F, no BLE skip)
Write RXPIPE_CFG ← 0x01 (Pipe 0 only)
Write PIPE0_RXADDR0-3 ← 0wnAddr bytes [0..3]
Write RF_CHANNEL ← 0x55 (calibration channel)
Write RF_DATARATE ← DataRate
Write 0x43 ← 0x3A, 0x44 ← 0x8C (RF analog tuning / TX 9 dBm)
Write 0x55-0x61, 0x66, 0x68, 0x6E ← (RF analog tuning table)
```

Step 6 – RF calibration (Page 1)

```
Write PAGE_CFG ← 0x01
Write CAL_CTL ← 0x08 (start VCO cal)
Poll 0x70 bit[6] == 1 (VCO done)
Write CAL_CTL ← 0x10 (start thermal cal)
delay 55 ms (mandatory)
Write STATE_CFG ← 0x76 (enter RX for freq cal)
delay 200 µs
Write CAL_CTL ← 0x20 (start freq cal)
Poll 0x7F bit[7] == 1 (freq done)
```

```

Write CAL_CTL    ← 0x40      (start phase cal 1)
Poll 0x6D bit[7] == 1      (phase 1 done)
Write CAL_CTL    ← 0x80      (start phase cal 2)
Poll 0x7F bit[7] == 1      (phase 2 done)
Write CAL_CTL    ← 0x00      (stop cal)
Write STATE_CFG  ← 0x74      (back to STB3)
Write PAGE_CFG   ← 0x00

```

Step 7 – Final setup

```

Write RF_CHANNEL ← OperatingChannel
Write RFIRQFLG   ← 0xFF      (clear all interrupts)
Write STATE_CFG  ← 0x76      (enter RX)

```

TX Operation

```

Write STATE_CFG ← 0x74      (STB3 – required before FIFO/address access)
Write TXADDR0-3 ← dst[0..3] (destination address, 4 bytes LE)
Write TXPLEN_CFG ← len(payload)
Write TRX_FIFO  ← payload[0..n-1]
Write RFIRQFLG  ← 0xFF      (clear stale flags)
Write STATE_CFG ← 0x75      (enter TX)
Poll RFIRQFLG bit[7] (TX_IRQ) (max ~5000 iterations)
Write RFIRQFLG  ← 0xFF
Write STATE_CFG ← 0x74      (STB3)
Write RFIRQFLG  ← 0xFF
Write STATE_CFG ← 0x76      (re-enter RX)

```

With `TRXMODE_CFG = 0x41` (single TX), the chip automatically returns to STB3 after transmitting.

RX Operation

With `RX_G00N=1` (`WMODE_CFG1` bit 7) the chip remains in RX after each received packet:

```

Poll RFIRQFLG bit[0] (RX_IRQ) – non-blocking check
If set:
  (Optional) Read STATUS3 (0x77) for dynamic payload length
  (Optional) Read STATUS0 (0x74) bits[6:4] for pipe number
  (Optional) Read STATUS1 (0x75) for received header byte 0
  Read TRX_FIFO ← payload (n separate transactions)
  Write RFIRQFLG ← 0x01      (clear RX_IRQ only)

```

TX Power Configuration

All power-setting registers span both pages. Switch to Page 1, write registers, switch back to Page 0, write Page 0 registers.

Register	Page	0 dBm	9 dBm
0x27	1	0xAA	0xAA
0x3C	1	0x13	0x17
0x46	1	0xBD	0xB0
0x48	1	0x88	0x88
0x43	0	0x3A	0x3A
0x44	0	0x84	0x8C

Carrier-Wave Test Mode

Used for regulatory testing (continuous unmodulated carrier).

```
Write PAGE_CFG ← 0x01
Write 0x41 ← 0x20
Write 0x42 ← 0x4E
Write PAGE_CFG ← 0x00
Write TRXMODE_CFG ← 0x81 (continuous TX)
Write STATE_CFG ← 0x75 (TX)
```

Exit:

```
Write PAGE_CFG ← 0x01
Write 0x41 ← 0x00
Write 0x42 ← 0x00
Write PAGE_CFG ← 0x00
Write STATE_CFG ← 0x74 (STB3)
Write TRXMODE_CFG ← 0x41 (restore single TX)
```

Sleep / Wakeup (09_sleep example)

```
Enter sleep:
  Write STATE_CFG ← 0x74 (STB3)
  Write STATE_CFG ← 0x21 (Sleep)

Exit sleep:
  Write STATE_CFG ← 0x22 (Wake)
  Write STATE_CFG ← 0x74 (STB3)
  delay 1 ms (crystal stabilisation)
  [resume normal TX/RX]
```

Register state is preserved during sleep. The crystal oscillator stops; the 1 ms delay after wakeup allows it to restabilise before RF operation.

BLE Mode Configuration Differences

When `WORK_MODE = 11` (BLE), the following registers differ from XN297L-compatible mode:

Register	XN297L	BLE	Notes
WMODE_CFG0	0x89	0xFC	3B CRC, BLE mode, whitening, little-endian
WMODE_CFG1	0xA2	0xB2	DPY_EN=1 (auto-length from PDU)
WHITEN_CFG	0x7F	0x53 / 0x33 / 0x73	BLE channel seed (ch 37/38/39); SKIP_ADDR=0 in BLE mode
PKT_EXT_CFG	0x00	0x60	Auto-insert 2 header bytes
TXHDR0_CFG	0x00	0x42	ADV_NONCONN_IND TxAdd=1
BLEMATCH_CFG0	0x00	0x04	Length filter = equal
BLEMATCHSTART_CFG	0x07	0x00	Filter from byte 0
RF_DATARATE_CFG	any	0x55	BLE requires 1 Mbps
0x5D (Page 0) ✱	0xDC	0xD4	RX demod tuning
0x61 (Page 0) ✱	0x2E	0x3E	RX demod tuning
0x6F (Page 0) ✱	0x00	0x10	PID_LOW_SEL=1 in BLE RX

In BLE TX: FIFO payload = AdvA (6 bytes LSB-first) + AdvData only. Header (PDU type + length) auto-inserted by chip.